Optimizing ARIMA and SARIMA Parameters for Seasonal Time Series Forecasting

A Data-Driven Approach to Determining the Optimal (p, d, q) and (P, D, Q, s) Parameters

BY: RAMSHA KHAN

BSE INSTITUTE
DATA SCIENCE

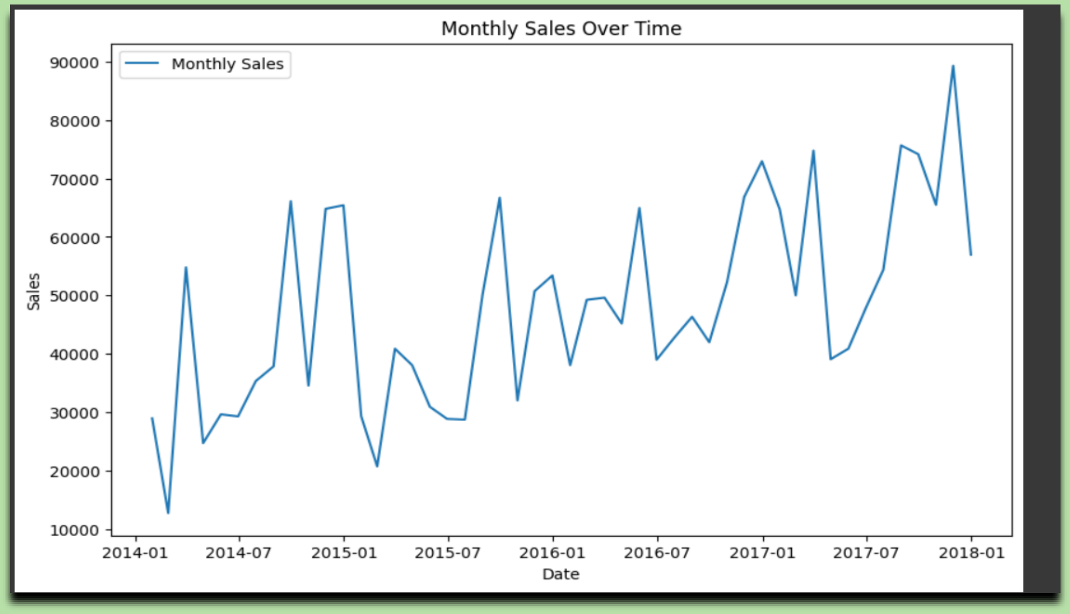
FINDING THE OPTIMIZED VALUE OF P,D,Q FOR ARIMA & SARIMA FOR SEASONAL DATA

- P stands for Autoregressive (AR) term. P in ARIMA/SARIMA represents the autoregressive (AR) term, which means the model uses the past p time steps' values to predict the current value, capturing the temporal dependence in the series.

- D stands for Differencing order. D in ARIMA/SARIMA is the seasonal differencing order, which means the model subtracts the value from its previous seasonal period to remove seasonal trends and make the series stationary.
- Q stands for Moving Average (MA) term, Q in SARIMA represents the seasonal moving average (SMA) term, which means the model uses the errors (residuals) from past seasonal periods to correct the current forecast, capturing seasonal noise patterns.

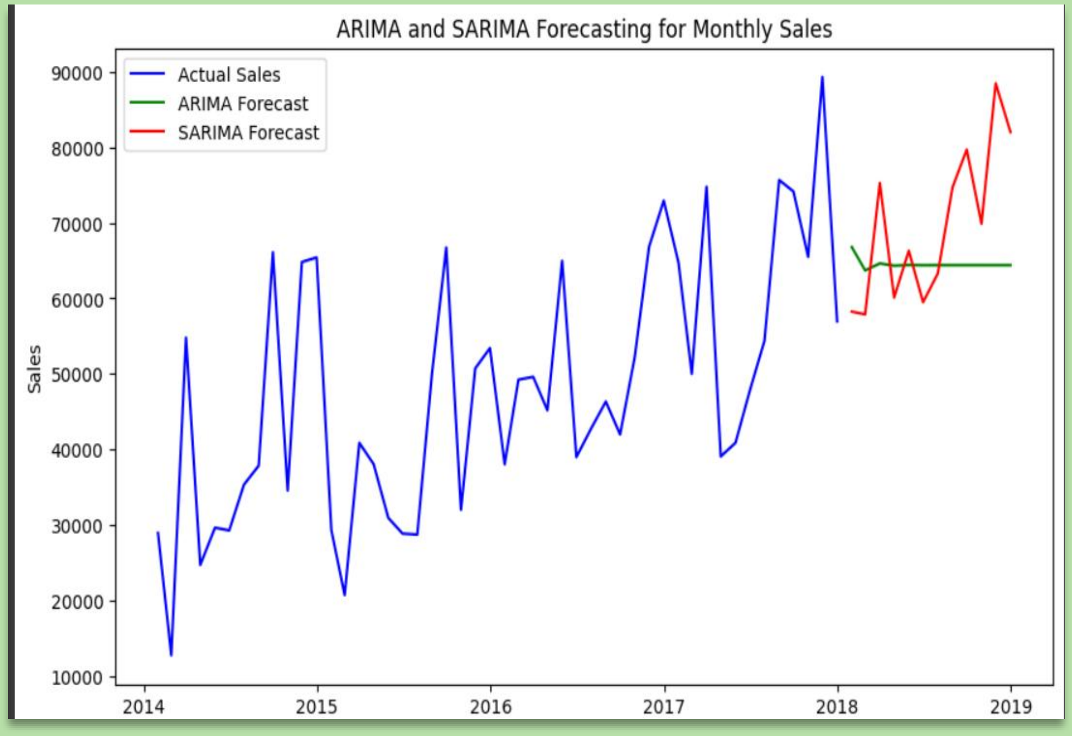
NO	COMBINATION (P , D, Q)
1	**(1, 1, 1)**
2	**(1, 1, 2)**
3	**(1, 2, 1)**
4	**(1, 2, 2)**
5	**(2, 1, 1)**
6	**(2, 1, 2)**
7	**(2, 2, 1)**
8	**(2, 2, 2)**

Data Graph

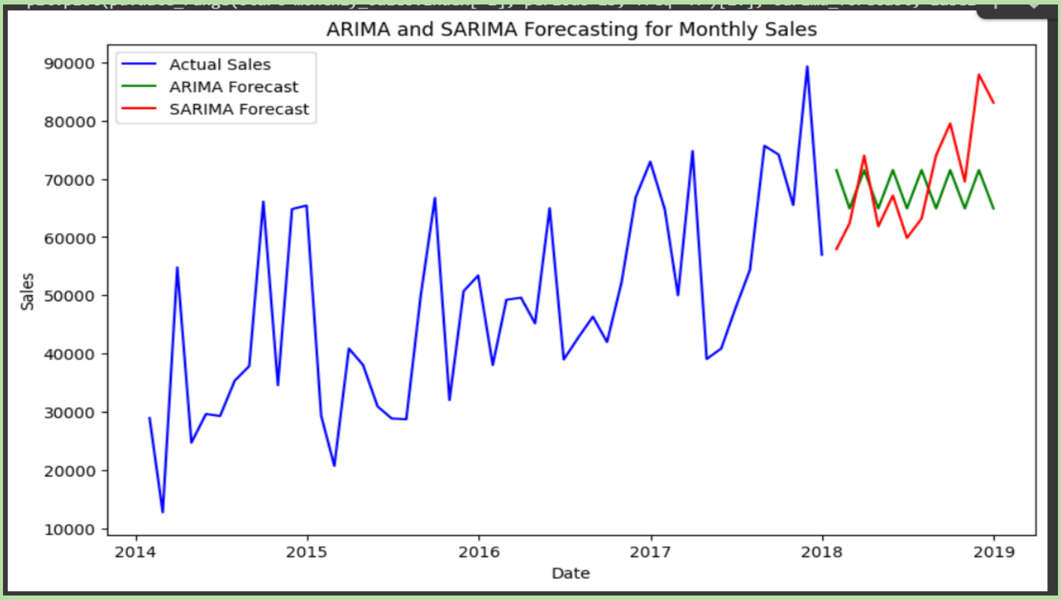


GRAPH

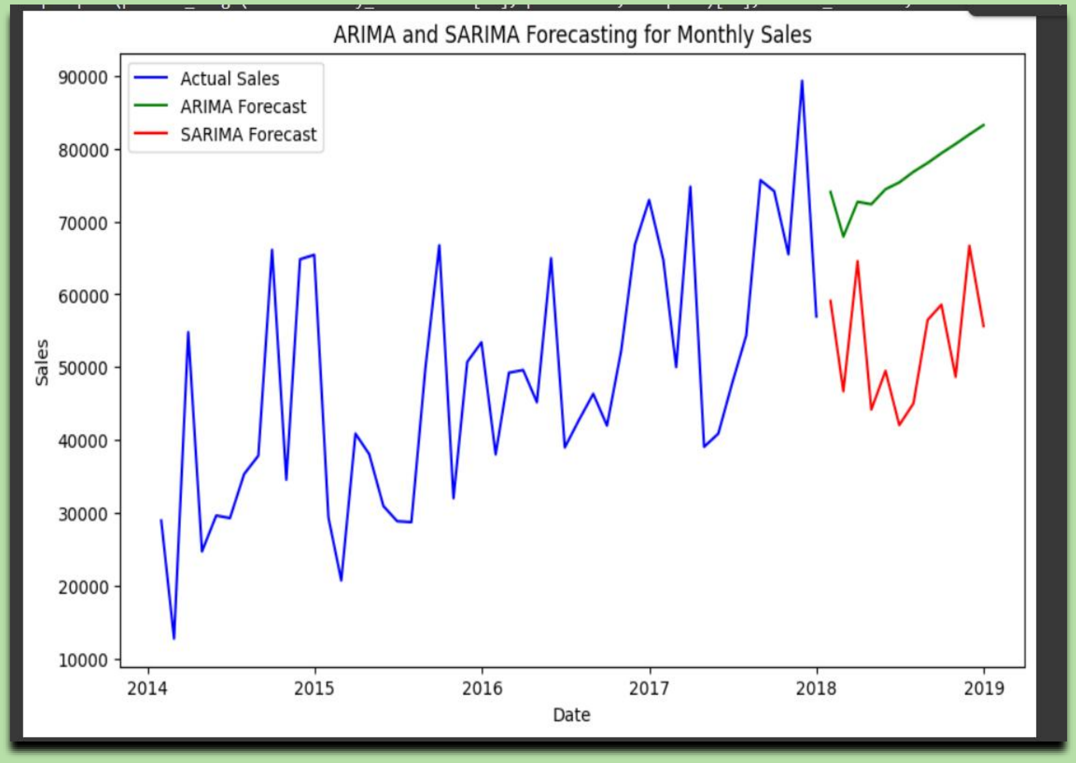
COMBINATION 1 -1,1,1



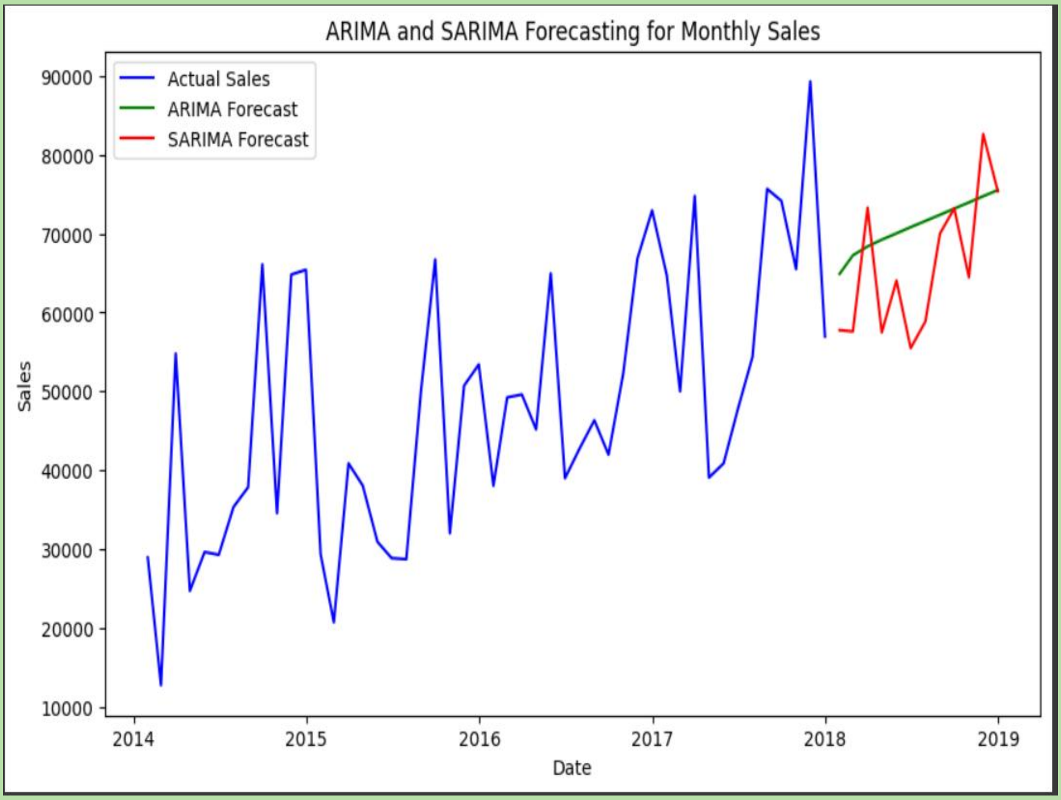
COMBINATION 2 - 1,1,2



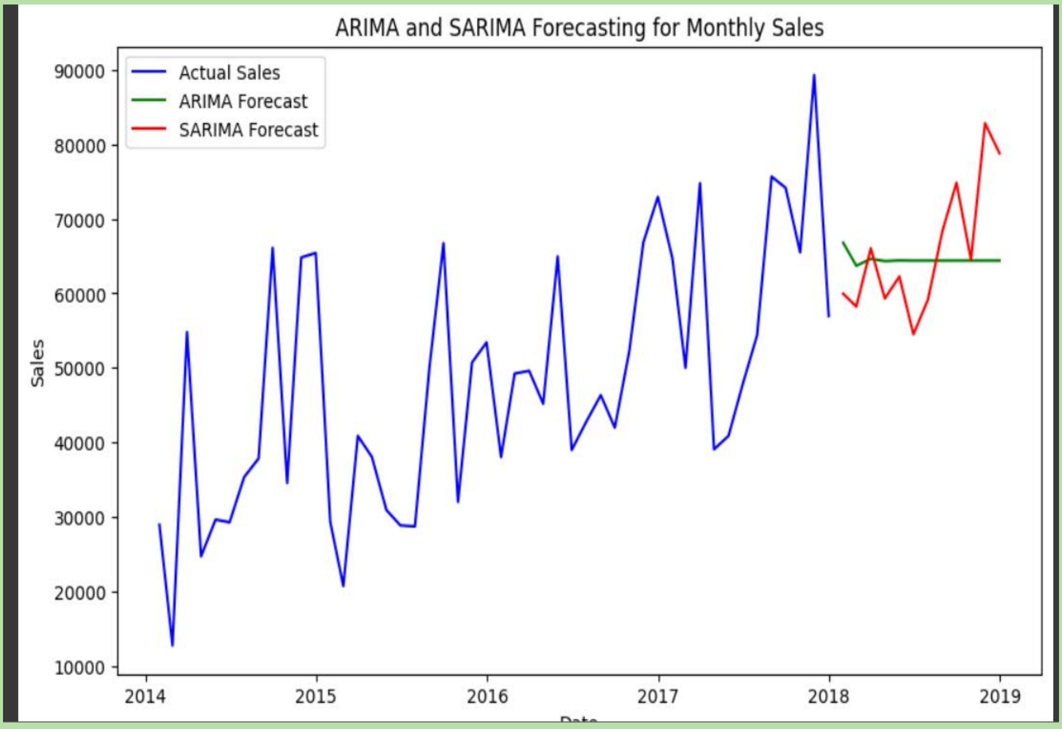
COMBINATION 3 - 1,2,1



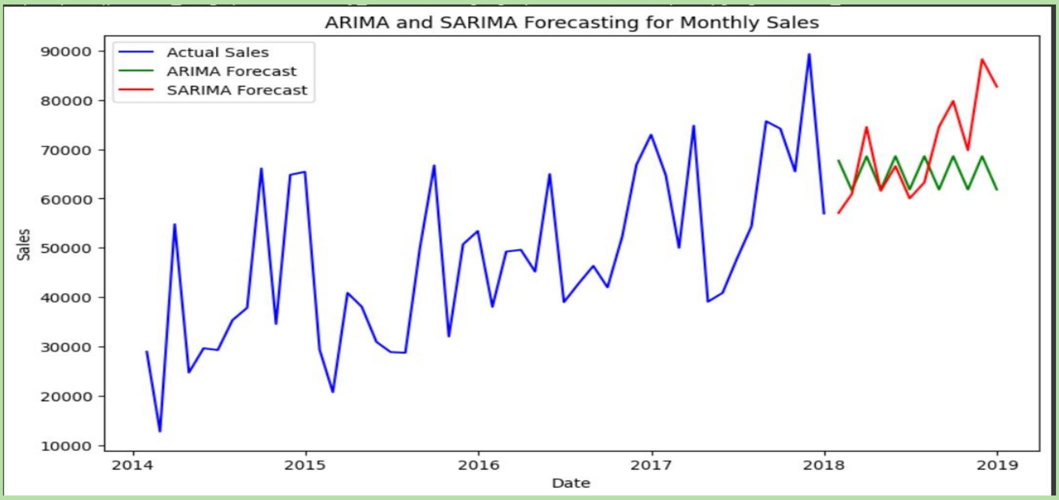
COMBINATION 4 - 1,2,2



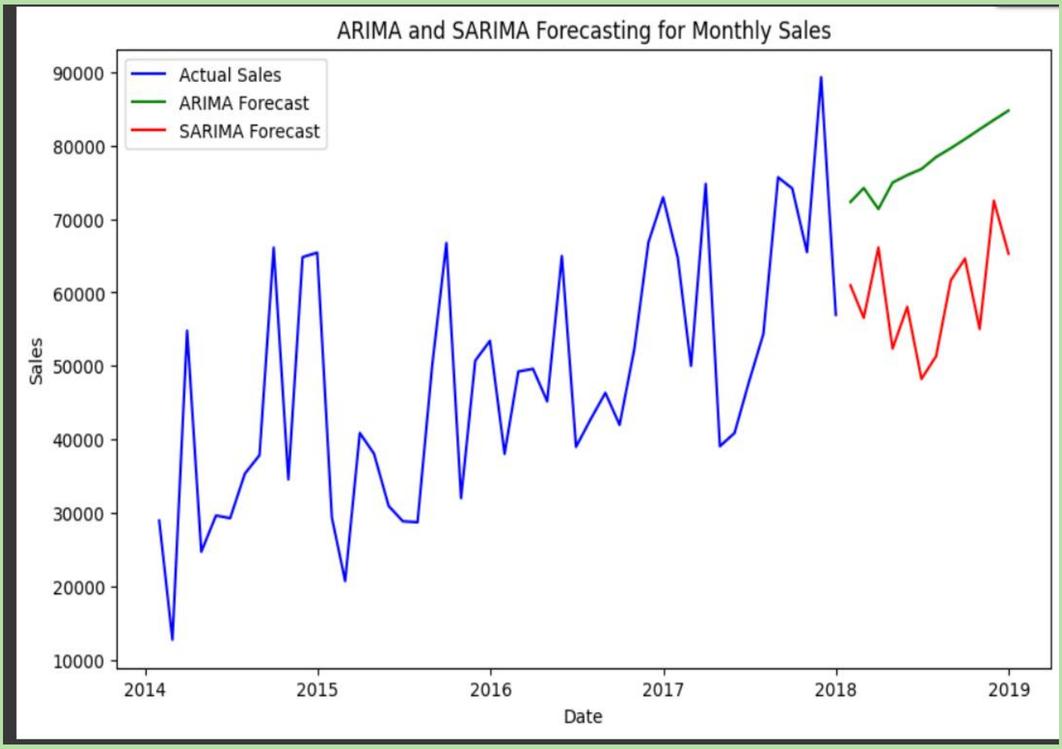
COMBINATION 5– 2,1,1



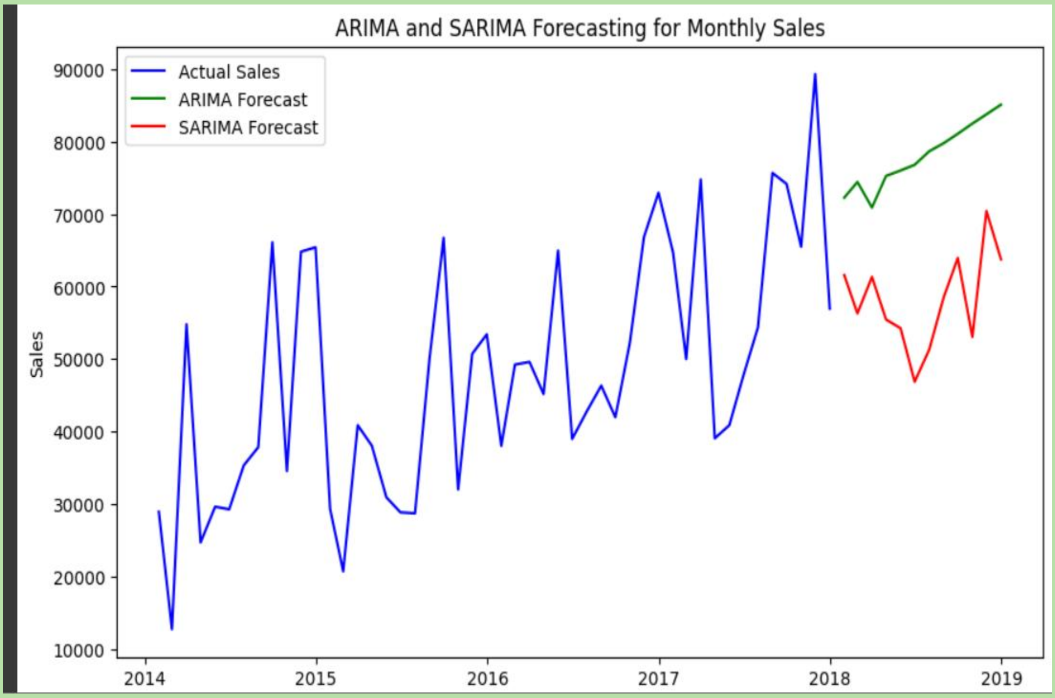
COMBINATION 6 - 2,1,2



COMBINATION 7 -2,2,1



COMBINATION 8 - 2,2,2



CONCLUSION

- ✓ **This project highlights the power of parameter tuning in ARIMA and SARIMA models for accurate seasonal time series forecasting. By systematically testing combinations of (p, d, q) and analyzing their results, we demonstrated how data-driven modeling can enhance predictive accuracy. This approach not only strengthens forecasting performance but also builds a strong foundation for real-world time series applications.**